

Roll No.

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Total No. of Pages : 02

Total No. of Questions : 16

**BCA (2014 to 2018)/B.Sc. (IT) (2015 & Onward)
(Sem.-2)**

OOPS USING C++

Subject Code : BSBC/BSIT-203

M.Code : 10052

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A** is **COMPULSORY** consisting of **TEN** questions **carrying TWO marks** each.
2. **SECTION-B** contains **SIX** questions carrying **TEN** marks each and students have to attempt any **FOUR** questions.

SECTION-A

Answer briefly :

1. What are unions?
2. Describe the use of constructors.
3. What are the advantages of Inheritance?
4. What is the use of Header files in C++?
5. What are the different types of Arrays in C++?
6. What is the use of Scope Resolution operator?
7. What is the significance of Operator overloading?
8. How are structures initialized? Explain with an example.
9. How can a protected member be made available for inheritance?
10. Describe the concept of objects and classes.

SECTION-B

11. a) How is C++ different from C language? What are the **object oriented features of C++**?
b) What is the difference between Structure and Union? Give **suitable examples to justify your answer**.
12. a) How are objects passed as function arguments? What is the **difference between Pass by value and Pass by reference**?
b) Give suitable program to describe the idea behind **Friend functions**.
13. a) Explain the concept of Inheritance. What are the uses of **Inheritance? What are its different types?**
b) What do you mean by Constructor Overloading? Give suitable **program to explain the concept**.
14. Explain the concept of Polymorphism with the help of an **example. What are the advantages of polymorphism?**
15. Write brief notes on :
 - a) Data hiding and Encapsulation
 - b) Use of Pure Virtual Functions
16. a) List the visibility modes used in defining derived classes in **Inheritance**.
b) Write a program to illustrate Function overloading concept.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

SECTION-B

2. Differentiate between C and C++.
3. Explain call by value, call by reference and call by address with examples.
4. Write a program to overload ++ and -- operators to increase and decrease the value of the object data member.
5. Explain the various types of access specifiers used in C++
6. What are the different types of inheritance in C++? Explain with an example.
7. Write a program to copy one file into another.

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Total No. of Pages: 02

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SECTION-A

1. Answer the following:
 - a) What are Structures?
 - b) Describe the use of destructors.
 - c) What are the advantages of Data hiding?
 - d) What is the use of Header files in C++?
 - e) What are the different types of Arrays in C++?
 - f) What do you mean by Late binding?
 - g) What is the significance of Pure virtual functions?
 - h) Write the syntax of declaring a function in a class?
 - i) How can a private member be made available for inheritance?
 - j) Describe the concept of Pass by value.

SECTION-B

2. a) What are the benefits of Object Oriented Programming?
b) Write a program to calculate the sum of all odd numbers between 1 and 50.
3. a) What is a Constructor? Explain its various types with examples.
b) Explain the concept of the Input stream and output stream with an example.
4. a) What do you mean by Inheritance? Write the difference between Multiple and Multilevel inheritance with the help of suitable examples.
b) How does function overriding differ from function overloading?
5. Explain the concept of Polymorphism with the help of an example. What are the advantages of polymorphism?
6. Write brief notes on:
 - a) Scope Resolution Operator
 - b) Use of Friend Functions
7. a) Use of Unions in C++
b) Write a program to illustrate the Binary and Unary Operator overloading concept.

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